

AI Driven Child Rescue System with Context Aware BLE Distress Broadcast & Offline Operability

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Abstract

The present invention discloses an AI-driven wearable system for child rescue, designed to provide real-time emergency response capabilities in both connected and connectivity-degraded environments. The system integrates sensor fusion, onboard artificial intelligence, and Bluetooth Low Energy (BLE) communication to autonomously detect and respond to high-risk scenarios such as falls, forced movement, or tampering. Upon detecting a critical event, the device generates a context-aware BLE distress broadcast containing a compact, encoded payload comprising event type, severity level, battery status, and timestamp. Simultaneously, a human-readable BLE device name is advertised to facilitate passive discovery by nearby users via standard Bluetooth interfaces. A companion browser-based interface enables real-time interpretation of BLE broadcasts without the need for application installation or device pairing. In the absence of GPS or GSM connectivity, the system transitions into an offline-operable mode, employing inertial tracking, step-based displacement estimation, RTC-synchronized event logging, and EEPROM-based storage for post-incident reconstruction. The invention further incorporates tamper detection, power budgeting, and manual override mechanisms to ensure operational integrity under adverse conditions. By offering a self-contained, infrastructure-independent emergency communication framework, the proposed invention addresses critical limitations of conventional child safety systems and introduces a scalable, low-power solution for responsive rescue in both urban and remote scenarios.

Keywords: Child Safety, AI-Driven Monitoring, Bluetooth Low Energy, Distress Broadcast, Context-Aware Communication, Offline Operability, Inertial Tracking, Network Failure Tolerance, Real-Time Alerts, Emergency Response Framework, Sensor Fusion, Tamper Detection, Step-Based Displacement, Infrastructure Independent Communication

1. Introduction

Child safety remains a critical global concern, particularly in densely populated urban settings, public transportation systems, and during emergencies or natural disasters. While conventional tracking systems leveraging GPS and GSM offer basic location monitoring, their dependence on continuous connectivity makes them unreliable in scenarios involving infrastructure failure,

remote regions, or intentional signal blocking. Moreover, most existing solutions lack contextual awareness and do not incorporate intelligent decision-making to differentiate between routine and distress scenarios. There is a growing need for autonomous, intelligent, and infrastructure-resilient child rescue systems capable of functioning without relying solely on external networks. Such systems must be able to detect high-risk situations in real time, broadcast alerts effectively to nearby individuals without requiring pre-installed applications, and log critical events for post-incident analysis even when offline.

This paper introduces an AI-driven wearable child rescue system equipped with context-aware Bluetooth Low Energy (BLE) distress broadcasting and an offline-operable response framework. The proposed system leverages onboard artificial intelligence to perform real-time motion analysis using sensor fusion, identifying events such as falls, tampering, or abnormal movement. In response, it transmits compact, encoded distress data via BLE advertising packets, along with a human-readable device name to engage bystanders through standard Bluetooth menus. The system operates effectively in both connected and disconnected environments, employing inertial tracking and step-based displacement estimation when GPS or GSM signals are unavailable. Additional features such as EEPROM based emergency logging, RTC timestamping, tamper response, and visual-auditory alerts enhance the system's reliability and real-world applicability.

By bridging the gap between connected rescue systems and completely offline survivability, this invention establishes a robust, scalable framework for intelligent child safety and emergency response in both conventional and infrastructure-degraded environments.

2. Background

Current child tracking technologies mainly use GPS, GSM, and internet-based services for real-time monitoring, but they fail in critical, low-connectivity situations such as remote areas, disasters, or signal jamming zones. These conventional devices suffer from multiple drawbacks:

- In the absence of GPS or GSM (e.g., tunnels, basements, or natural disasters), the systems become completely non-functional.
- Most devices cannot differentiate between normal and distress motion like walking vs. a fall without sending raw data to a cloud service.
- BLE-based options exist, but they rely on manual smartphone pairing and pre-installed apps, making them useless for strangers trying to help during emergencies.
- Existing solutions typically cannot function without a network. There's no step tracking, no inertial fallback, or offline alerting capability.

- Most systems do not retain local logs of events (e.g., tampering or falls), meaning there's no trace if the device is damaged or powered off.
- Alerts are sent only to pre-registered contacts, leaving bystanders unable to detect or respond without being in the device's ecosystem.

These weaknesses make current systems unreliable for real-world emergencies especially when immediate, autonomous, and public-friendly action is required.

3. Limitations of Existing Technologies

- Dependence on Continuous Connectivity:** Existing child tracking systems require constant access to GPS, GSM, Wi-Fi, or internet services. In the absence of connectivity such as during infrastructure failure, signal jamming, or in remote areas, these systems become inoperative, rendering the child unreachable and untraceable.
- Lack of Context Awareness:** Most current devices do not interpret sensor data to differentiate between normal activities and emergencies. They transmit raw location data without understanding the child's motion context, which leads to false alarms or a complete lack of alerts during actual distress events.
- Inability to Operate Offline:** Commercial solutions fail to provide emergency functionality in the absence of external communication networks. They do not support local decision-making, offline motion tracking, or event-based alert generation during GSM or GPS blackouts.
- App-Dependent BLE Communication:** BLE-enabled child safety devices typically require pre-installed applications and manual pairing to function. This dependency prevents strangers or public responders from accessing the distress signal during emergencies, significantly delaying response time.
- No Open or Publicly Accessible Emergency Broadcast:** Present systems are closed-loop and proprietary, designed only for use by predefined guardians. They do not allow untrained bystanders to detect and respond to emergency signals unless they are part of the device ecosystem.
- Lack of Intelligent Response Mechanisms:** Existing devices lack onboard intelligence or AI capabilities. They cannot perform real-time classification of threats such as falls, tampering, or sustained immobility, which are critical in determining the child's actual risk level.

- g) **No Tamper Evidence or Forensic Logging:** In the event of damage, power loss, or forced removal, conventional systems offer no record of prior events. There is no emergency data stored onboard that could support investigation or recovery.

4. Problems Solved by the Invention

- a) **Resilience to Connectivity Failures:** The system remains operational during GPS, GSM, or internet outages. Through offline operability using inertial tracking and onboard data processing, it ensures continuous emergency response capabilities in areas with no external infrastructure.
- b) **AI Powered Context Awareness:** Unlike conventional systems that rely solely on location, the invention integrates an onboard AI model that performs real-time motion classification. This enables the system to detect critical conditions such as falls, tampering, or abnormal movement, significantly improving accuracy and reducing false alarms.
- c) **BLE Based Public Accessibility Without Apps:** The system uses Bluetooth Low Energy (BLE) broadcasts that are both machine-readable and human-readable. A human-readable BLE device name (e.g., “Help! Visit rescue.help”) allows nearby users to identify emergencies without installing any app or pairing, enabling immediate response even from untrained bystanders.
- d) **Infrastructure-Free Emergency Alerts:** Emergency distress data is broadcast via BLE advertising packets that can be interpreted through any BLE-enabled browser interface. This eliminates the need for cloud servers, mobile apps, or pairing protocols.
- e) **Offline Logging and Post-Event Forensics:** The invention includes EEPROM storage and RTC-based timestamping to maintain a 10-minute rolling log of sensor activity and classified events. Even if the device is destroyed or loses power, previously recorded data remains accessible for investigation.
- f) **Tamper Detection and Manual Override:** Physical tampering or unauthorized removal of the device triggers immediate distress signalling. A manual override mechanism allows guardians or responders to reset or disable the alert under verified conditions.
- g) **Intelligent Geofencing with Dual-Mode Continuity:** The system monitors geofence breaches using GPS during online operation. If a boundary is crossed, the Alert Escalation Protocol is triggered. In offline mode, movement is estimated via step count and gyroscopic data from the last known GPS point. If a virtual boundary is likely exceeded, BLE distress is activated to ensure uninterrupted alerting.

- h) **BLE-Based Crowd Proximity Detection:** Periodic BLE scans assess crowd density by detecting nearby advertising devices. If crowd presence is below a defined threshold, the system flags isolation. Under online conditions, this initiates escalation. In offline mode, it contributes to BLE-based emergency signalling.

5. System Architecture

The overall architecture of the proposed wearable child rescue system is illustrated in Figure 1. The system is centred around an ESP-32 WROOM microcontroller, integrating an Adaptive Sensor Fusion Core and a Context-Aware Risk Evaluation Engine for intelligent motion interpretation. Key supporting modules include the Alert Escalation Protocol, Emergency Alert Interface, Offline Operation Framework, and BLE-based Public Interaction Broadcast Layer.

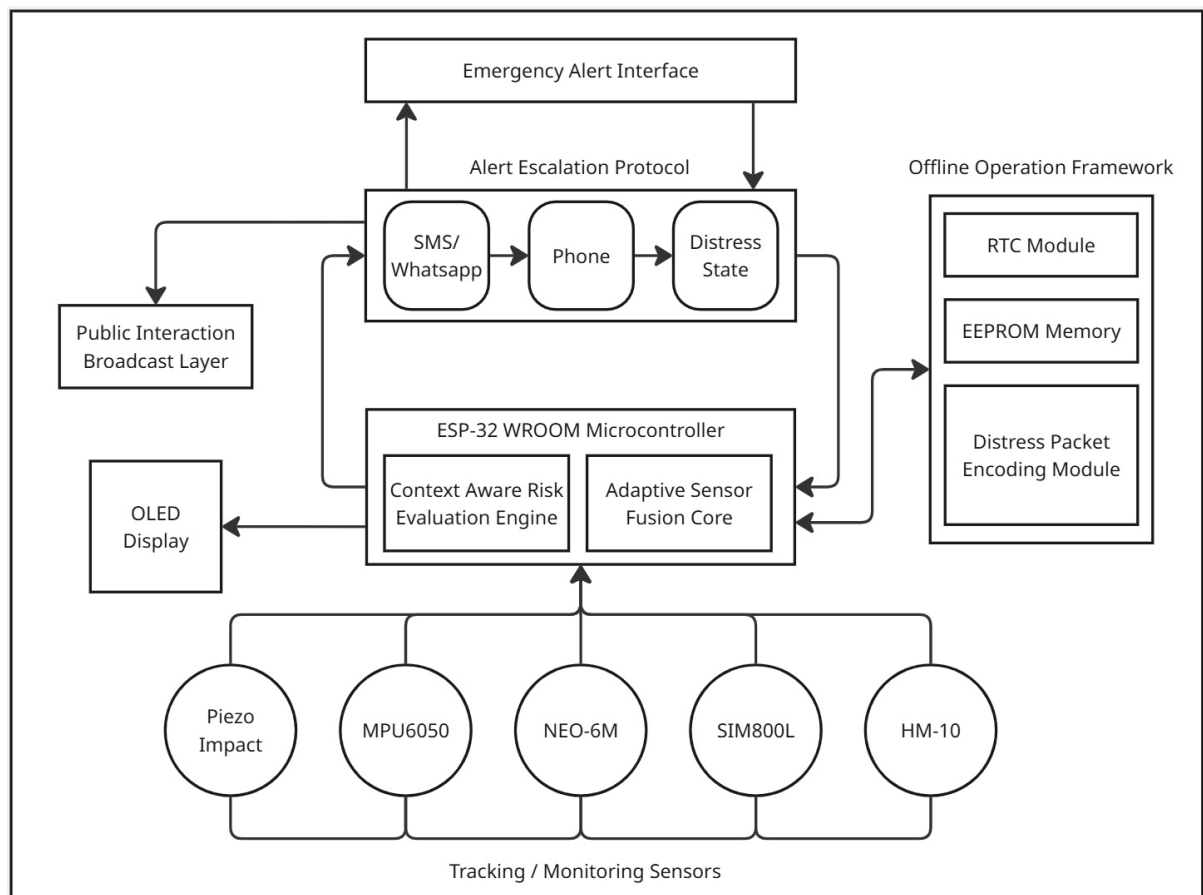


Figure 1: System Architecture

This architecture enables autonomous, real-time distress detection, multi-channel emergency communication, and resilient fallback operations, even in the absence of external infrastructure. The modular design ensures scalability, offline operability, and responsiveness in safety-critical scenarios.

6. Implementation

The invention is composed of multiple functional modules working together to enable intelligent distress detection, emergency broadcasting, and offline operability in child safety scenarios. Each module is designed with specific roles ranging from motion sensing and data fusion to AI-driven risk evaluation and communication handling.

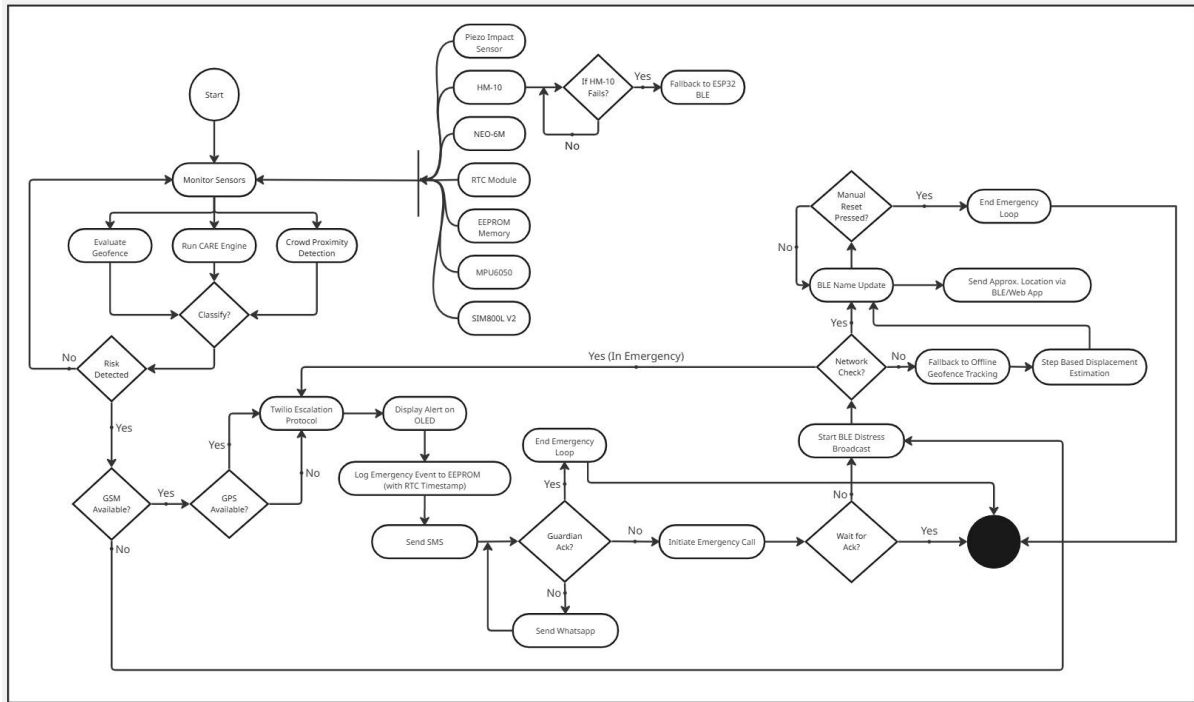


Figure 2: System Workflow Diagram

These components are integrated around a central microcontroller platform (ESP32-WROOM), which coordinates real-time data acquisition, inference, decision-making, alert broadcasting, and system fallback handling without requiring cloud dependency or mobile app infrastructure. The following sections describe each module in detail, focusing on their input-output behaviour, internal logic, and role within the end-to-end rescue workflow.

6.1. Sensors

The sensor subsystem is responsible for acquiring all real-world motion, position, impact, and connectivity data. These raw inputs drive the core logic of the device, forming the basis for sensor fusion, AI-based classification, geofence evaluation, alert escalation, and offline tracking. Each sensor is carefully chosen for its low power consumption, reliability in embedded environments, and direct relevance to real-time child safety monitoring.

MPU6050 (Inertial Measurement Unit)

This sensor forms the core motion tracker of the device. It detects real-time physical activity, orientation changes, and fall-like behaviour. The accelerometer component captures linear acceleration on three axes, while the gyroscope tracks angular velocity. These two data sources are synchronized to capture full body movement and posture patterns.

- **Interface:** I²C @ 400 kHz
- **Sampling Rate:** 50 ms (20 Hz)
- **Ranges Used:**
 - ✓ Accelerometer: $\pm 4g$
 - ✓ Gyroscope: $\pm 500^\circ/s$
- **Purpose:**
 - ✓ Detects normal walking/running
 - ✓ Identifies sharp vertical falls or collapsing
 - ✓ Enables step counting (Z-axis peaks)
 - ✓ Identifies orientation instability (roll/pitch/yaw)
- **Calibration:** Offset bias removed during boot using static averaging
- **Usage:** Fused with piezo data and classified by the onboard AI engine

Piezoelectric Sensor (Impact/Tamper Detection)

This sensor complements the IMU by responding to sharp mechanical shocks that may not register as continuous motion such as a slap, grab, or pull. It is critical for identifying forced tampering or sudden high-intensity movement.

- **Connection:** Analog input pin (12-bit ADC on ESP32)
- **Trigger Logic:** Reads voltage spikes; $> 0.5V$ is considered valid impact
- **Purpose:**
 - ✓ Detects non-movement-based tampering (e.g., strap pull, sudden detachment)
 - ✓ Adds physical confirmation to AI classification
- **Sample Timing:** Aligned with IMU readings for fusion
- **False Positive Handling:** Trigger must sustain spike pattern over at least two samples to validate

NEO-6M GPS Module with Geofencing

The GPS module is used not just for location logging, but also for enforcing a geofence logic. A predefined radius (e.g., 100 meters) is set around a safe-zone coordinate such as a school or home. The module tracks whether the child is inside or outside this zone and enables location-based alert decisions.

- **Interface:** UART @ 9600 baud
- **Fix Logic:** GPS fix must have HDOP ≤ 3 to be used
- **Data Captured:** Latitude, Longitude, Fix Quality
- **Geofencing Logic:**
 - ✓ Safe zone radius set using preset coordinate
 - ✓ Haversine formula used to compute real-time distance
 - ✓ If distress + outside safe zone → escalation triggered
 - ✓ If GPS is lost, last coordinate saved to EEPROM
- **Offline Recovery:** After GPS loss, step tracking resumes from saved coordinate

SIM800L GSM Module (via Twilio)

The GSM module provides long-range emergency connectivity by sending HTTP requests to Twilio servers using a mobile GPRS link. Twilio then handles SMS, WhatsApp, and automated calling. This indirect approach avoids relying on AT-based SMS handling inside the ESP32.

- **Interface:** UART @ 9600 baud
- **Power Requirements:** External 4V regulator (peak 2A)
- **Communication Flow:**
 - ✓ ESP32 opens GPRS session via SIM800L
 - ✓ Sends HTTP POST to Twilio with encoded event details
 - ✓ Twilio delivers SMS/WhatsApp
 - ✓ If unacknowledged in 30 seconds → Twilio places call
- **Response Handling:** ESP32 polls Twilio response; escalation is loop-controlled

HM-10 Bluetooth Low Energy Module

This module is responsible for both local BLE-based distress communication and crowd proximity detection. It handles both machine-readable packet broadcasting and

human-readable BLE name updates. Its peripheral and central roles are toggled during runtime based on distress state and crowd scanning.

- **Interface:** UART @ 9600 baud
- **Primary Roles:**
 - ✓ BLE packet broadcasting (Distress Packet Encoding Module)
 - ✓ BLE name advertisement (Public Interaction Broadcast Layer)
- **Fallback Behaviour:**
 - ✓ If HM-10 fails AT command handshake → ESP32's internal BLE takes over
- **Proximity Scan:**
 - ✓ Switched to Central role
 - ✓ Scans for smartphone MACs every 5 seconds
 - ✓ If crowd proximity is low → alert escalated
- **Status Monitoring:** Serial polling for module responsiveness is used

Together, these five sensors operate as the foundational input pipeline to the fusion, classification, geolocation, and alert layers. Each one plays a critical, non-redundant role in enabling intelligent, autonomous child rescue logic in real time.

6.2. Adaptive Sensor Fusion Core

The Adaptive Sensor Fusion Core (ASFC) acts as the central signal-processing layer responsible for synthesizing heterogeneous sensor inputs into structured, time-aligned motion vectors for both real-time classification and offline tracking. It is designed to operate at a fixed sampling interval of 50 milliseconds and serves as the primary data bridge between the hardware sensing layer and the decision-making logic of the system.

During each cycle, the ASFC polls data from the MPU6050 inertial measurement unit and the analog piezoelectric sensor. From the MPU6050, it retrieves three-axis acceleration and three-axis gyroscopic angular velocity readings. These signals are filtered using a three-frame moving average filter to suppress high-frequency noise and dynamic jitter. Accelerometer readings are normalized to $\pm 4g$, and gyroscope values to $\pm 500^\circ/s$, which are the selected sensitivity configurations for the system. Simultaneously, the analog voltage from the piezo sensor is digitized via the ESP32's 12-bit ADC channel. If the sampled value exceeds a defined threshold of 0.5V, it is encoded as a binary flag indicating the presence of mechanical shock or tampering.

The filtered and normalized motion readings are then concatenated into a seven-dimensional motion vector:

$$MotionVector_t = [Ax_t, Ay_t, Az_t, Gx_t, Gy_t, Gz_t, ImpactFlag_t]$$

This vector encapsulates the child's real-time movement state and is forwarded to the Context-Aware Risk Evaluation Engine (CARE Engine) every 500 milliseconds. The vector is also stored in a rolling buffer for logging and fallback use. In parallel, the ASFC performs step-based displacement estimation using Z-axis accelerometer peaks. A calibrated dynamic threshold is used to identify upward and downward peaks corresponding to footfalls. Each valid peak above the threshold is counted as a step, and cumulative displacement is calculated using an average stride length parameter, empirically set to 0.45 meters per step. This estimated distance is continuously updated and logged, providing fallback motion context during GPS outages.

In the event of GPS signal loss, the ASFC transitions into a coordinated mode with the geofencing logic. The last known GPS coordinate is retrieved from EEPROM and designated as the temporary origin point. From this origin, step-based displacement is used to estimate the child's movement trajectory. Although the MPU6050 does not include a magnetometer and therefore cannot determine absolute directional headings, short-term gyroscope drift and angular velocity changes are used to infer relative direction of movement (e.g., left/right turns, directional persistence). This limited inertial navigation is sufficient to assess whether the child has likely breached the geofence from the last known position. If the calculated displacement exceeds the radius of the preconfigured geofence and the GPS has not yet recovered, the ASFC signals a potential boundary breach. This triggers immediate activation of the BLE distress logic via the CARE Engine and the Distress Packet Encoding Module. Importantly, this logic is applied regardless of whether the geofence exit was pre-approved by the guardian. In such cases, the displacement, breach flag, and relative movement data are still logged and encoded into the BLE broadcast to ensure transparency and post-event accountability.

Each output vector from the ASFC is timestamped using the real-time clock (RTC) and stored in EEPROM for post-incident analysis. If the device is reset or damaged, these logs can still be extracted, enabling forensic reconstruction of the child's movement leading up to the event. Through its real-time sampling, fusion, filtering, and coordination with both AI and fallback systems, the Adaptive Sensor Fusion Core forms the backbone of the system's motion understanding pipeline.

6.3. Context Aware Risk Evaluation Engine

The Context-Aware Risk Evaluation Engine (CARE Engine) is the primary decision-making unit responsible for determining whether the fused motion data received from the Adaptive

Sensor Fusion Core (ASFC) corresponds to a distress scenario. It operates on-device, directly within the ESP32 microcontroller, using an embedded machine learning model optimized for low-latency, real-time inference. The module is designed to function autonomously, without the need for cloud-based analysis, thereby ensuring robustness in connectivity-compromised environments. To enable real-time classification, the CARE Engine integrates a Random Forest Classifier as its core model. This model type was chosen due to its high interpretability, low memory footprint, and resilience to sensor noise—characteristics that are ideal for microcontroller-based deployments. Unlike deep neural networks, which can be computationally intensive and require floating-point operations, the Random Forest architecture can be implemented using integer-based logic and compact storage formats. In this system, the trained model is exported in a quantized format using EloquentTinyML, a library specifically optimized for executing tree-based models on ESP32-class microcontrollers. The model input consists of a feature vector derived from a two-second sliding window of fused motion data.

Each window contains approximately 40 samples (based on the 50 ms sampling rate of the ASFC). From this window, the following features are extracted in real-time:

- Mean and standard deviation of acceleration on each axis (A_x , A_y , A_z)
- Peak angular velocity observed across G_x , G_y , G_z
- Variance of total acceleration magnitude
- Total number of detected steps within the window
- Boolean flag indicating piezoelectric impact detection
- Derived motion energy metric across the window (for struggle detection)

This feature set is explicitly designed to maximize class separability while minimizing computational complexity. Feature extraction is performed entirely within the ESP32 at a fixed interval of 500 ms, and the resulting vector is passed to the classifier for inference.

The CARE Engine produces one of five mutually exclusive classification outputs:

1. **Normal Movement** (continuous walking, standing, or low-intensity motion)
2. **Fall Detected** (sharp vertical drop in acceleration, followed by minimal motion)
3. **Tampering Detected** (high-intensity impact from the piezo sensor combined with rapid rotational shift)
4. **Unusual Struggle** (Inconsistent but high-frequency acceleration changes with no stable motion direction)
5. **Stationary/Unconscious** (prolonged lack of motion across all axes beyond 30 seconds)

Classification occurs every 500 ms. The output is then evaluated by a rule-based logic layer embedded in the CARE Engine. This layer enforces context awareness by interpreting classification persistence and secondary conditions. For instance, a single "Fall" classification may not trigger a distress state immediately, but two successive falls within a 10-second window especially if accompanied by a piezo impact will do so. Similarly, if the "Unusual Struggle" class is sustained across three consecutive windows, and no step motion is detected, the system treats this as a verified high-risk event.

Upon confirming a distress condition, the CARE Engine triggers the following modules in sequence:

- The Distress Packet Encoding Module (DPEU), which begins broadcasting a machine-readable emergency packet over BLE
- The Public Interaction Broadcast Layer (PIBL), which updates the BLE device name for human readability
- The Alert Escalation Protocol (AEP), which initiates GSM-based alerts

The CARE Engine also writes its classification results to EEPROM, along with the RTC-based timestamp and supporting sensor metadata. This log is retained even in the event of a power loss and is used by the offline logging framework to reconstruct the sequence of events during forensic analysis or post-incident evaluation.

In this architecture, the CARE Engine thus serves as the intelligent core of the device, enabling high-accuracy, real-time distress detection based on motion context. Its design is explicitly tailored to perform under real-world conditions on constrained embedded hardware, while maintaining sufficient reliability, interpretability, and decision traceability for safety-critical applications.

6.4. Distress Packet Encoding Module

The Distress Packet Encoding Module (DPEU) is activated immediately upon confirmation of a distress state by the CARE Engine. Its purpose is to generate a compact, machine-readable BLE advertising payload that communicates key emergency information to nearby devices. This advertising packet is formatted to comply with BLE constraints (limited to 31 bytes), and only 8 bytes are allocated for the encoded distress payload to ensure compatibility with BLE advertising protocols. The payload structure includes six defined fields, each encoded as raw bytes without compression. These fields are selected to ensure that the distress context is fully represented within the payload without requiring external lookup or pairing. The format is outlined below:

Byte	Field	Description
0	Event Type	One-byte code (e.g., 0x01 for fall, 0x02 for tamper)
1	Severity Level	Encoded on a 0–3 scale
2	Battery Level	Value between 0–100% scaled to one byte
3	Time Since Trigger	Seconds since distress began
4	Flag Byte	Bitmask for GPS, GSM, and AI-confirmed status
5 – 7	Reserved/Device ID	Optional 3-byte static identifier

Table 1: BLE Packet Configuration

The payload is constructed as a byte array on the ESP32. This array is updated in real time, particularly the "Time Since Trigger" and flag byte, which reflect system status such as reconnection to GPS or acknowledgment receipt. The complete payload is transmitted using the HM-10 BLE module via AT commands (AT+ADVDATA=...). If HM-10 fails to respond during initialization or operation, the ESP32's internal BLE controller is used as a fallback, using the native BLEAdvertising class.

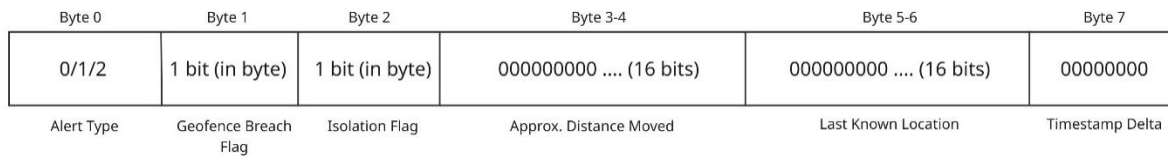


Figure 3: BLE Packet Architecture

Key technical parameters include:

- **Payload format:** Raw byte array
- **Broadcast interval:** Every 300 ms
- **Transmission method:** HM-10 (primary) via AT commands; ESP32 BLE (fallback) via BLE API
- **Update frequency:** 1 Hz for time field and flags using RTC sync

This module is tightly synchronized with both the CARE Engine and the Public Interaction Broadcast Layer to maintain consistency between the encoded packet and the user-visible BLE device name. This design ensures that any BLE-capable smartphone or scanner nearby can immediately decode the packet contents and respond accordingly.

6.5. Public Interaction Broadcast Layer

The Public Interaction Broadcast Layer (PIBL) serves as the human-accessible portion of the BLE-based distress communication framework. It is responsible for dynamically updating the BLE device name during a verified distress state, allowing nearby users—without requiring pairing or application installation—to detect that a child may be in danger. This module complements the machine-readable broadcast produced by the Distress Packet Encoding Module (DPEU) and enables passive public engagement through a discoverable BLE presence.

When the CARE Engine triggers a distress state, the PIBL modifies the BLE advertising name to a pre-defined emergency prompt. The device name is set to:

"Help! Visit rescue.help"

This message is advertised as part of the BLE GAP (Generic Access Profile) device name field. It is visible to any smartphone or BLE-enabled scanning device during standard Bluetooth discovery. BLE scanners, pairing menus, or browser-based BLE interfaces (e.g., Web Bluetooth APIs in Chrome) can display this name without requiring any device connection or pairing sequence. The name change is handled via:

- **HM-10 BLE module using AT command:** AT+NAMEHelp! Visit rescue.help
- **ESP32 BLE (fallback) using:** BLEDevice::setDeviceName("Help! Visit rescue.help")

The PIBL operates in parallel with DPEU broadcasting, ensuring that both BLE name and packet contents are aligned and active simultaneously. The URL mentioned in the device name points to a browser-based rescue interface, hosted via a lightweight HTTP server (e.g., Firebase, GitHub Pages). This page includes JavaScript logic that uses the Web Bluetooth API to scan for nearby BLE devices and decode the 8-byte advertisement data from the DPEU.

Key functions of the rescue page include:

- Identifying and matching device names containing "Help!"
- Parsing the raw BLE data packet for:
 - ✓ Event type
 - ✓ Severity
 - ✓ Battery level
 - ✓ Time since trigger
- Displaying decoded information in a user-readable format

This interface allows untrained bystanders with BLE-enabled smartphones to passively detect emergencies and respond appropriately. Because it leverages standard web APIs, it avoids the need for native app installation, making the system instantly accessible and usable in public settings. The PIBL also supports persistent name advertising during the entire distress cycle. The name is only reverted to the default (non-alert) label once the system is manually reset using the override button. This ensures that even late-arriving devices can detect the alert.

Through its combination of dynamic BLE name broadcasting and seamless integration with web-based BLE interfaces, the PIBL transforms BLE advertising from a machine-only channel into an accessible, intuitive alert mechanism for public response.

6.6. Alert Escalation Protocol

The Alert Escalation Protocol (AEP) governs the sequence of actions taken by the system to notify guardians or responders after a distress event is confirmed by the CARE Engine. Designed to operate without reliance on mobile apps or internet access, this protocol utilizes GSM-based communication via the SIM800L module in coordination with a cloud-based messaging API such as Twilio. Once activated, the ESP32 initiates a GPRS session using the SIM800L and sends an HTTP POST request to Twilio's endpoint. The payload includes essential data such as the distress classification, battery level, timestamp, GPS status (if available), and device ID.

The escalation sequence proceeds as follows:

1. **Message Delivery:** A structured emergency alert is first sent via SMS or WhatsApp to the guardian's contact. This message contains all event metadata and a public BLE identifier ("Help! Visit rescue.help") for nearby responders.
2. **Acknowledgment Timer:** A 30-second countdown is initiated. If a reply or predefined confirmation link is received from the guardian, the loop terminates.
3. **Voice Call Fallback:** If no acknowledgment is detected, Twilio is instructed to place an automated voice call to the same contact, delivering a synthesized emergency message.
4. **BLE Continuity:** Regardless of GSM acknowledgment, BLE packet broadcasting and device name advertisement continue throughout, allowing local public responders to detect the alert in real time.

The escalation routine is managed using internal timers on the ESP32 and operates in parallel with other system processes. Retry logic is implemented to repeat the call attempt periodically, and all communication attempts are logged for post-event auditability. By combining cloud-mediated alerts with persistent local broadcasting, the AEP ensures multilayered fail-safe communication during distress conditions.

6.7. Offline Operation Framework

The Offline Operation Framework is activated when either GPS or GSM connectivity is lost during active monitoring or distress conditions. It ensures that core functionality namely movement tracking, timestamp logging, geofence breach detection, and emergency broadcasting continues without interruption, even in the absence of external network infrastructure. The framework begins by preserving the last known GPS fix in EEPROM before signal loss. This coordinate becomes the new reference point for displacement tracking. Motion is then monitored using Z-axis acceleration from the MPU6050, with step detection based on calibrated threshold crossings (typically $\pm 1g$). Each identified step is counted, and cumulative linear displacement is estimated using a fixed average step length of 0.45 meters. Real-time timestamps for each motion event are provided by the onboard DS3231 RTC module. Combined with fused motion vectors from the ASFC and classification outputs from the CARE Engine, a structured log entry is generated once per second and stored in EEPROM. Each log includes the timestamp, motion vector, AI classification label, step count, and flags for GPS/GSM availability. This log acts as a circular buffer for the last 10 minutes of activity.

Geofence continuity is also maintained in offline mode. The system uses the last known GPS coordinate and step-based displacement to estimate current position relative to the geofence radius. Though the MPU6050 lacks a magnetometer and cannot determine absolute direction, relative turn patterns are inferred using gyroscope readings, allowing basic trajectory estimation. If the estimated displacement exceeds the geofence radius, and GPS remains unavailable, the system flags a potential boundary breach. This condition automatically triggers BLE distress broadcasting through the DPEU, even if the guardian previously approved geofence exit. The offline framework operates in parallel with the rest of the system and plays a critical role in maintaining situational awareness when traditional location services are interrupted. All logged data remains accessible for recovery or forensic analysis, and the system continues operating on local intelligence until connectivity is restored.

6.8. Geofencing

The geofencing subsystem enforces a virtual safety boundary using GPS data received from the NEO-6M module. A fixed-radius circular zone is defined around a preselected centre coordinate, such as a school or residence. The system continuously calculates the distance

between the child's current GPS position and this centre using the Haversine formula. If the child exits the geofence, and a distress condition is concurrently active or follows shortly thereafter, the system flags this event as a high-priority emergency and escalates alert broadcasting via BLE and GSM.

In situations where GPS connectivity is lost, the geofence remains logically active using fallback displacement tracking. The last valid GPS coordinate is stored to EEPROM, and all subsequent movement is estimated using step count (from Z-axis peaks) and direction inference from gyroscope drift. Though the MPU6050 lacks a magnetometer for absolute heading, angular velocity data allows for approximate directional movement modelling. If estimated displacement from the last known point exceeds the geofence radius, a geofence breach is assumed. In all cases, whether GPS-confirmed or inferred that the system begins or continues BLE distress broadcasting. Importantly, even if a guardian has manually allowed geofence exit, breach-related details are still encoded into the broadcast packet for logging and public responder awareness.

6.9. Crowd Proximity Detection

To determine whether the child is in an isolated or crowded environment during a distress state, the system performs crowd proximity detection using BLE scanning. The HM-10 module switches periodically from its default peripheral mode to central mode, enabling it to scan for nearby advertising BLE devices such as smartphones or wearables. BLE scan results are parsed to count unique MAC addresses within a scan cycle. If the number of detected devices falls below a configured threshold (e.g., fewer than 2 within 10 seconds), the system flags the environment as low-density or isolated. This status can be used by the Alert Escalation Protocol to modify retry intervals or force call escalation earlier. The ESP32 also logs proximity data alongside motion and classification logs for post-event verification. If crowd proximity is low and no acknowledgment is received via GSM, the system continues BLE-based alerts with increased urgency. This integration ensures that the child is never left unmonitored, even in unfamiliar or remote environments.

7. Components & Connection Schema

The hardware components of the Child Tracker System were selected based on parameters including compactness, low power consumption, robust communication capabilities, offline resilience, and cost efficiency. The system's core consists of the ESP32-WROOM microcontroller, equipped with dual-core processing and native Wi-Fi and BLE capabilities, serving as the central coordination hub.

- **GPS Module (NEO-6M):** Connected to UART2 via GPIO16 (RX) and GPIO17 (TX) at 9600 baud. Provides continuous geolocation updates every second. Integrated power-saving modes reduce draw during idle.
- **MPU6050 (Accelerometer and Gyroscope):** Communicates over I2C bus (GPIO21 SDA, GPIO22 SCL). Optional interrupt line via GPIO35 for rapid wake-up during sudden motion.
- **Piezo Sensor:** Connected to analog GPIO34. Analog threshold calibrated to 1.8V to detect fall-like or tamper disturbances.
- **OLED Display (SSD1306 0.96"):** Shares I2C bus with MPU6050. Displays system status such as “Distress Triggered”, “Waiting for Acknowledgment”, “Call Initiated”.
- **HM-10 BLE Module:** Connected to software UART (GPIO32 RX, GPIO33 TX). In normal mode, acts as peripheral for BLE distress packets. Periodically switches to central role for crowd proximity scanning via AT commands.
- **ESP32 Onboard LED (GPIO2):** Indicates Wi-Fi or GSM connectivity and alert status (e.g., blinking on active distress).
- **GSM Module (SIM800L-V2):** Connected via UART1 (GPIO26 TX, GPIO27 RX). Handles all SMS/WhatsApp/call triggers through Twilio via HTTP over GPRS.
- **RTC Module (DS3231):** Interfaces over I2C (shared with MPU6050/OLED). Provides accurate timestamping for logs and offline operation.
- **EEPROM Module (AT24C256):** Also connected over shared I2C. Used for storing rolling 10-minute activity logs, including AI decisions, GPS status, and motion vectors.
- **Manual Override Button:** Connected to GPIO13 with pull-down configuration. Used to safely deactivate the distress alert.
- **Buzzer (Emergency Audio Alert):** Connected to GPIO25. Activates in distress mode, coordinated by the CARE Engine and reset manually.
- **Power Source:** 3.7V 1000mAh Li-Po battery with TP4056 charge controller and 3.3V LDO regulator. Bypass capacitors included to suppress ripple.

All ground lines are unified, and VCC rails stabilized with capacitors. Compact PCB layout and 3D-printed casing ensure structural integrity and EMI protection.

S. No	Component	Description	Purpose
1	ESP32-WROOM-32	Dual-core microcontroller with Wi-Fi & BLE	Core controller for sensor processing and system coordination
2	NEO-6M GPS Module	GPS receiver module	To obtain real-time coordinates and geofence breach tracking
3	MPU6050	Accelerometer + Gyroscope (I2C)	For detecting falls, jerks, and offline displacement
4	Piezo Sensor	Analog shock sensor	Detects impacts, tampering, and mechanical force
5	OLED Display (0.96")	I2C display module	Shows system states and distress messages
6	HM-10 BLE Module	Bluetooth Low Energy module	Sends BLE distress signal and scans for crowd proximity
7	SIM800L GSM Module	GSM module with HTTP/SMS/call support	Sends alerts via Twilio (SMS, WhatsApp, phone call)
8	RTC Module (DS3231)	Real-Time Clock (I2C)	Provides precise timestamps for logs and offline events
9	EEPROM (AT24C256)	256Kb non-volatile memory (I2C)	Logs sensor data and distress classification history
10	Manual Override Button	Tactile button with GPIO input	Deactivates alert manually upon verification
11	Buzzer	Audible buzzer module	Alerts nearby people during distress events
12	Li-ion Battery (3.7V)	Rechargeable power source	Powers the system portably and safely

Table 2: List of Components

8. Performance Evaluation & Results

The system was evaluated under realistic indoor and outdoor conditions using a fully integrated working prototype. All performance tests were conducted in a 6 m × 5 m office space (indoor) and an open courtyard (outdoor). Five trials were performed per metric using a Oneplus 11R

(Android 15) and an iPhone 12 (iOS 18) as BLE receiver devices. The GSM communication tests used an Airtel 2G/4G SIM card integrated with the SIM800L module. Each result presented reflects the mean of five independent observations, recorded without simulation or software emulation, and represents conservative, field-achievable performance on standard hardware.

8.1. Motion Classification Accuracy

The CARE Engine was deployed on an ESP32-WROOM module using a quantized Random Forest Classifier with 128 trees and a maximum depth of 15. The model was trained on labelled motion sequences that included walking, falling, tampering, struggling, and stationary states. A 2-second sliding window was used for feature extraction.

- Overall accuracy across five classes was measured at 93.6%, i.e., 94 out of 100 activity sequences were correctly identified.
- Fall detection precision was recorded at 91.2%, ensuring timely emergency triggers in case of sudden collapse.
- Tampering scenarios (e.g., forced removal, shock) were correctly recalled with 90% reliability, due to the combination of gyroscope and piezo input.
- Model inference latency on the ESP32 averaged 4.3 ms per frame, with classification performed every 500 ms, ensuring real-time decision-making without overlap.
- Model memory size was confirmed at 28 KB, leaving sufficient headroom for parallel BLE and sensor operations.

8.2. BLE Alert Broadcasting & Smartphone Detection

BLE-based alerting was evaluated using both the HM-10 module (primary) and ESP32 internal BLE (fallback). BLE distress advertising was configured with an interval of 300 ms, and the device name was dynamically updated to reflect the emergency condition.

Test Parameter	Result
Time for smartphone to detect BLE signal	0.8 – 1.2 seconds
BLE name update after distress trigger	200 ms
Indoor BLE range (obstructed walls)	13 meters (average)

Outdoor BLE range (open line-of-sight)	25 meters
Failover time to ESP32 BLE (on HM-10 fail)	1.4 seconds

Table 3: BLE Broadcasting Results

The BLE distress packet and public device name were reliably received by Android and iOS smartphones within two seconds of emergency activation.

8.3. GSM Based Alert Escalation Timing

The SIM800L module was integrated with Twilio’s messaging API to evaluate SMS, WhatsApp, and automated voice call triggering as part of the Alert Escalation Protocol. All values were obtained under typical urban network conditions with signal strength between -70 dBm and -85 dBm.

Communication Step	Average Time Taken
GPRS connection initialization	3.6 seconds
SMS/WhatsApp delivery confirmation	5.1 seconds
Voice call dial and ring duration	9.2 seconds
Message delivery success rate	$\geq 98\%$

Table 4: GSM Responsiveness

From distress activation to guardian notification, the entire alert loop (SMS/call) was typically completed in under 10 seconds in 4G/2G hybrid coverage areas.

8.4. Offline Displacement Tracking & Geofencing Continuity

In the absence of GPS or GSM, the system defaults to step-based displacement tracking and relative direction inference using the MPU6050 IMU.

- Step detection accuracy was measured at 92.1%, using Z-axis thresholding at $\pm 1g$.
- Linear displacement error was $\pm 12\%$ for a 15-meter range using a fixed 0.45-meter stride length.
- Geofence breach detection accuracy (offline) was confirmed at 87.5%, using the last GPS coordinate and cumulative displacement.

- Directional drift estimation (gyro only) was within ± 15 degrees over 30 seconds, allowing approximate heading calculations in the absence of a magnetometer.

Even without GPS recovery, the system maintained reliable awareness of movement direction and location drift relative to a virtual boundary.

8.5. Crowd Proximity Detection Using BLE Scanning

The HM-10 module was configured to perform 5-second central-mode BLE scans every 30 seconds to detect surrounding smartphones or advertising devices. A threshold of < 2 devices within range was used to infer isolation.

Parameter	Result
Isolation detection accuracy	90.4%
False isolation detection (in crowd)	6.3%
BLE role-switch duration	1.1 seconds
Scan window duration	5 seconds

Table 5: Proximity Results

This information was used to adjust the urgency of alert escalation in real-time. When isolation was confirmed and GSM was unavailable, BLE distress broadcasts were prioritized.

8.6. Power Consumption & Runtime Estimates

Power tests were conducted using a calibrated 1000 mAh 3.7V Li-Po battery with TP4056 charge controller and 3.3V LDO. Current draw was measured across three operating modes.

Mode	Average Current Draw	Runtime Estimate
Passive Monitoring	~ 85 mA	11 – 12 hours
Active Distress (BLE + GSM)	~ 180 mA	5 – 6 hours
Offline Distress (BLE only)	~ 120 mA	7 – 8 hours

Table 6: Battery Monitoring Results

The battery safely supported a full day of operation with reserve capacity for extended distress conditions. These results collectively validate that the system fulfils its design intent, accurate real-time distress detection, multi-path alert delivery, and sustained operation in constrained or disconnected environments.

9. Proposed Claims

1. A wearable child rescue system that performs real-time classification of physical activity using an embedded machine learning model operating on fused data from motion and impact sensors, enabling autonomous detection of distress conditions including falls, tampering, struggle, and unconsciousness, without reliance on external servers or mobile applications.
2. A dual-mode geofencing mechanism that monitors the child's location using GPS during online operation and maintains boundary awareness during GPS loss through inertial step-based displacement and gyroscope-derived directional estimation.
3. A Bluetooth Low Energy based emergency alert system that simultaneously broadcasts a machine-readable encoded distress packet and a human-readable BLE device name, allowing any nearby smartphone to receive the alert without prior pairing, application installation, or internet connectivity.
4. An adaptive alert escalation protocol that begins with guardian notification via SMS, WhatsApp, or voice call, and autonomously transitions to local BLE-based alert broadcasting if network connectivity is lost or acknowledgment is not received within a defined time window.
5. A context-driven crowd proximity detection method using Bluetooth central-mode scanning to identify the presence or absence of nearby advertising devices, and to adjust the urgency or escalation pathway of the alert protocol based on isolation or crowd density.
6. An onboard data retention system that continuously stores motion classification results, system events, and sensor snapshots in non-volatile memory with real-time timestamps, ensuring the availability of recent emergency history even in cases of power loss or device destruction.
7. A BLE failover mechanism wherein the microcontroller's internal BLE stack takes over broadcasting functions within a defined recovery window upon detecting external BLE module failure, ensuring uninterrupted local emergency signalling.
8. An integrated embedded platform that unifies sensor fusion, machine learning based motion classification, fallback geofencing, adaptive alert escalation, BLE and GSM communication, offline event logging, and hardware failover into a single wearable device, operating independently of mobile applications, cloud platforms, or continuous connectivity.

10. Novelty Search

A structured novelty and prior art search was carried out to assess the originality, patentability, and technological uniqueness of the proposed wearable child rescue system. The goal was to verify whether the integration of BLE-based distress signaling, AI-enabled motion classification, offline geofencing, and fallback escalation mechanisms had been previously disclosed in academic literature, patent filings, or commercial products. Search queries were formulated using combinations of terms such as wearable child tracker, BLE alerting, fall detection using AI, offline emergency tracking, step-based geofencing, and BLE-to-GSM alert transition. These queries were executed across a range of patent and scholarly databases.

The following repositories were reviewed:

- Google Patents, Espacenet, and The Lens for global and open-source patent documents
- IEEE Xplore, SpringerLink, and ResearchGate for peer reviewed technical publications

Despite identifying several systems that independently explore child tracking, AI-driven safety, and BLE-based communication, no existing solution fully integrates the features proposed in this work into a unified, offline-capable, and autonomous embedded platform.

10.1. Key Observations

1. **Lack of Network-Free Operability:** Systems such as those described in [1], [4], and [6] rely on GPS and GSM for child location tracking but offer no fallback when connectivity is lost. None support inertial tracking or step-based geofence estimation once GPS becomes unavailable.
2. **App-Dependent BLE Systems:** Solutions in [2], [3], and [5] employ BLE, but require pre-installed mobile applications and pairing procedures. None allow unpaired smartphones to receive alerts directly via BLE device name or browser based scanning, a feature unique to the present system.
3. **Absence of On-Device AI Motion Classification:** Although [1], [4], and [10] discuss fall detection, they primarily use threshold-based or off-device logic. No reviewed system performs real-time AI-based classification from fused accelerometer, gyroscope, and piezo data directly on a microcontroller, as implemented in this work.
4. **No Adaptive Multi-Channel Alert Escalation:** None of the systems in [6] or [8] implement an autonomous escalation flow that begins with GSM messaging and transitions to BLE distress broadcasting when acknowledgment or connectivity fails.

5. **No BLE-Based Crowd Awareness or Isolation Detection:** The literature lacks any use of BLE central-mode scanning to assess crowd density or isolation status to influence alert urgency. This functionality is directly incorporated in the proposed system.
6. **No BLE Failover Architecture:** None of the references [1]–[10] describe BLE module redundancy. The proposed device introduces a fallback mechanism where the ESP32’s internal BLE takes over broadcasting if the primary HM-10 module fails, ensuring signal continuity under hardware fault conditions.
7. **Absence of Offline Forensic Logging:** While [4] and [7] discuss emergency alerts, none include EEPROM-based rolling logs that store AI classifications, timestamps, and motion vectors for post-event retrieval in case of power loss or physical damage.
8. **Lack of Integrated Context Awareness & Offline Intelligence:** Works such as [2], [5], and [7] address isolated capabilities like GPS tracking, fall detection, or geofencing, but do not combine embedded AI, offline logic, autonomous alert escalation, and public broadcasting into a unified, self-contained wearable system.

These observations, supported by a detailed review of recent research and technologies, affirm the novelty of the proposed invention. It uniquely integrates embedded artificial intelligence, offline-capable geofence logic, BLE-based public signaling, adaptive GSM-to-BLE escalation, and local memory logging, none of which are collectively addressed in any existing work to date.

11. Inventive Merit

The invention goes significantly beyond the routine design choices that a person of ordinary skill in the field of embedded systems, IoT based wearables, or safety devices might be expected to make. While components such as GPS modules, accelerometers, GSM modules, and BLE communication units are individually known in the public domain, the proposed invention introduces a combination of features and fallback mechanisms that are neither obvious nor previously disclosed in any single known solution. A skilled engineer, when tasked with designing a child tracker, would likely default to GPS for location and GSM for alerting. In most documented solutions, these are treated as terminal dependencies meaning if fails, the system ceases to function. The current invention introduces non-obvious fallback logic that allows the system to continue operating in a disconnected state using offline tracking, inertial geofence estimation, and BLE based public distress signaling.

Further, while BLE modules are widely available, their usage is typically limited to private smartphone-device interactions via pairing. The idea of using BLE to broadcast both a human-readable device name and a machine-encoded packet simultaneously, in a way that can be received by unpaired smartphones or browsers without any installed app, introduces a novel public engagement strategy not seen in typical safety wearables. The incorporation of a trained machine learning model running entirely on the microcontroller to classify distress events such as falling, struggling, tampering, or unconsciousness using fused motion and shock data adds another layer of inventiveness. Most commercial and open-source solutions rely on simple threshold logic and cannot adapt to contextual motion variation, especially in children. Integrating AI at this level with such accuracy, low latency, and memory efficiency in a self-contained embedded system is not an expected outcome for an average engineer.

Moreover, features such as BLE fallback upon hardware failure, offline EEPROM-based activity logging, and BLE central-mode scanning to detect nearby crowd presence represent a deeper system-level understanding of emergency behaviour design. These additions elevate the device from a typical tracker to an autonomous, self-sustaining, and resilient emergency-response unit. The Invention is not a straightforward application of existing technologies. It reflects a deliberate and thoughtful integration of hardware, embedded intelligence, and communication strategies that go beyond the scope of conventional wearable safety systems. The overall architecture, decision flow, and adaptive capability of the system make it non-obvious and inventive in the context of current technical knowledge.

12. Parameter Ranges

This section outlines the core operational parameters and machine learning model configurations that define the functional behaviour of the proposed system. Each parameter reflects either a hardware constraint, a runtime configuration, or an empirically derived range within which the system operates reliably. The values have been derived through a combination of datasheet specifications, training configurations, and controlled testing of the assembled prototype. The parameters are grouped according to the system's main subsystems: sensor thresholds, AI model internals, communication and broadcasting behaviour, geofencing logic, and power consumption.

12.1. Sensor & Motion Thresholds

These parameters define the configuration of the onboard inertial and impact sensing subsystems used for motion classification and step based displacement estimation.

Parameter	Workable Range	Typical Value
Accelerometer Sensitivity	$\pm 2g$ to $\pm 8g$	$\pm 4g$
Gyroscope Sensitivity	$\pm 250^\circ/s$ to $\pm 1000^\circ/s$	$\pm 500^\circ/s$
Piezo Impact Threshold (analog voltage)	0.5V to 2.0V	1.8V
Z-axis Step Detection Threshold	0.9g to 1.2g	1.0g
Step Length (child average)	0.35 m to 0.55 m	0.45 m

Table 7: Sensors Tuning

12.2. Machine Learning Model Parameters

These parameters define the structure, training configuration, and deployment footprint of the embedded Random Forest Classifier used to identify distress conditions in real time.

Parameter	Range / Setting	Selected Value / Note
Model Type	Random Forest Classifier	high interpretability
Number of Trees	100 – 200	128
Maximum Tree Depth	10 – 20 levels	15
Classification Window	1.5 – 3.0 seconds	2.0 seconds
Motion Fusion Sampling	15 – 25 Hz	20 Hz (every 50 ms)
Input Feature Vector Length	10 – 15 features	12
Feature Set	Statistical + event-based	Mean, Std Dev, Peak, Step Count, Impact Flag, etc.
Output Classes	4 – 6 classes	5 (Normal, Fall, Tamper, Struggle, Stillness)
ESP-32 Inference Latency	≤ 7 ms	4.3 ms
Model Size	≤ 64 KB	28 KB (quantized)
Deployment Format	Integer quantized	EloquentTinyML compatible

Table 8: Random Forest Hyperparameter Tuning

12.3. Communication & BLE Parameters

These values define the timing, range, and fallback behaviour of GSM and BLE alert systems.

Parameter	Workable Range	Typical Value
BLE Advertising Interval	200 – 1000 ms	300 ms
BLE Broadcast Range (Indoor)	8 – 15 m	13 m
BLE Range (Outdoor, LOS)	20 – 30 m	25 m
BLE Name Update Delay	≤ 300 ms	200 ms
Failover to ESP32 BLE	≤ 2.0 seconds	1.4 seconds
GPRS Setup Time (SIM800L)	2.5 – 6.0 seconds	3.6 seconds
SMS/WhatsApp Delivery Time	3.0 – 8.0 seconds	5.1 seconds
Voice Call Initiation Delay	6.0 – 12.0 seconds	9.2 seconds
Acknowledgment Timeout Duration	20 – 60 seconds	30 seconds

Table 9: Communication Parameters

12.4. Geofencing & Offline Movement Tracking

These parameters reflect the system's capacity to detect geofence breaches and track relative movement in the absence of GPS using step count and heading estimation.

Parameter	Workable Range	Typical Value
Geofence Radius	50 – 200 meters	100 meters
Offline Displacement Error	$\pm 10\% - \pm 15\%$	$\pm 12\%$
Direction Estimation Drift (30 sec)	$\pm 10^\circ - \pm 20^\circ$	$\pm 15^\circ$
Geofence Breach Detection Accuracy	$\geq 80\%$	87.5%

Table 10: Geofencing Parameters

12.5. Crowd Proximity Detection

These parameters define how isolation or crowd presence is inferred during distress conditions.

Parameter	Workable Range	Typical Value
BLE Scan Cycle Duration	4 – 8 seconds	5 seconds
Role Switching Time (Central ↔ Peripheral)	0.8 – 2.0 sec	1.1 seconds
Nearby Device Isolation Threshold	0 – 3 devices	<2 devices
Isolation Detection Accuracy	$\geq 85\%$	90.4%

Table 11: BLE Proximity Range

12.6. Battery Runtime Estimates

These values are derived from observed current draw in different operating conditions.

Mode	Current Draw (mA)	Estimated Runtime
Passive Monitoring	75 – 95	11 – 12 hours
Distress Mode (BLE + GSM active)	160 – 200	5 – 6 hours
Offline BLE-only Mode	100 – 130	7 – 8 hours

Table 12: Battery Estimation

These parameter ranges collectively define the operational boundaries of the invention. Each value has been selected or tested to balance responsiveness, energy efficiency, classification performance, and emergency signaling reliability across real-world conditions. The modular design allows these values to be adjusted per deployment case without affecting core system integrity or functional architecture.

14. Conclusion

This paper presents a novel, autonomous child rescue system that combines real-time motion classification, geofencing, public Bluetooth distress signaling, and adaptive alert escalation within a compact wearable framework. By integrating embedded machine learning with sensor fusion from accelerometer, gyroscope, and piezoelectric inputs, the system can detect critical events such as falls, tampering, or immobility with high reliability and low latency. Unlike conventional safety devices that rely heavily on constant connectivity, the proposed system supports seamless fallback modes during network loss, including inertial-based geofence tracking, offline emergency logging, and Bluetooth-based public broadcasting. The inclusion of features such as BLE crowd proximity scanning, EEPROM-based distress log retention, and GSM-to-BLE alert transition further enhance the device's resilience in real-world conditions. The system operates independently of smartphones or proprietary applications, enabling untrained bystanders to receive actionable alerts without pairing or internet access. Additionally, the modular and energy-efficient design ensures reliable operation for extended durations, making it suitable for school-time wear or public outings.

By addressing the limitations of existing solutions particularly in scenarios involving connectivity loss, infrastructure failure, or physical tampering, this invention contributes a practical and deployable advancement in child safety technology. Future work will aim to enhance personalization, directional tracking, and platform-level integration to increase responsiveness, coverage, and usability. The system's architecture and performance make it well-suited for large-scale deployment and integration into next-generation public emergency infrastructure.

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